PRACTICAL ASSIGNMENT

RESEARCH IMPACT & BIBLIOMETRICS

Course: Research Methodology

Topic: Author & Journal Metrics (h-index, i10-index, and Journal Impact Factor)

PART 1: AUTHOR METRICS

Task 1: Manual Calculation of Author Metrics

Q1. Calculate the h-index for Researcher A and Researcher B. Show the step-by-step process.

The h-index is the highest value h such that a researcher has at least h publications, each of which has received at least h citations.

Researcher A

The citation counts in descending order are 45, 28, 18, 12, 9, 6, 3, and 1.

Paper Rank	Citations	h-index condition
1	45	45 ≥ 1: Satisfied
2	28	28 ≥ 2: Satisfied
3	18	18 ≥ 3: Satisfied
4	12	12 ≥ 4: Satisfied
5	9	9 ≥ 5: Satisfied
6	6	6 ≥ 6: Satisfied
7	3	3 ≥ 7: Not satisfied
8	1	1 ≥ 8: Not satisfied

At rank 6, the sixth paper has 6 citations, so the condition is satisfied. At rank 7, the seventh paper has only 3 citations, which is less than 7. Therefore, the h-index of Researcher A is 6.

Researcher B

The citation counts in descending order are 120, 15, 11, 10, 8, and 2.

Paper Rank	Citations	h-index condition
1	120	120 ≥ 1: Satisfied
2	15	15 ≥ 2: Satisfied
3	11	11 ≥ 3: Satisfied
4	10	10 ≥ 4: Satisfied
5	8	8 ≥ 5: Satisfied

6	2	$2 \geq 6$: Not satisfied

At rank 5, the fifth paper has 8 citations, so the condition is satisfied. At rank 6, the sixth paper has only 2 citations, which is less than 6. Therefore, the h-index of Researcher B is 5.

Q2. Calculate the i10-index for both researchers.

The i10-index is the number of publications that have received at least 10 citations.

Researcher A: Four papers have at least 10 citations (45, 28, 18, and 12). Therefore, the i10-index of Researcher A is 4.

Researcher B: Four papers have at least 10 citations (120, 15, 11, and 10). Therefore, the i10-index of Researcher B is 4.

Q3. Compare both profiles.

Researcher A shows more consistent productivity across multiple papers, with 8 publications and an h-index of 6. Researcher B has the higher peak impact on a single paper, which has 120 citations, but has a slightly lower h-index of 5. Both researchers have the same i10-index of 4.

PART 2: JOURNAL IMPACT FACTOR (JIF)

Task 2: Journal Metric Calculation

Q1. Write the standard formula for calculating the 2-Year Journal Impact Factor (JIF).

Two-Year Journal Impact Factor (JIF) = (Citations received in the evaluation year to citable items published during the previous two years) ÷ (Total number of citable items published during those two years).

For the 2025 evaluation year:
JIF_{2025} = Citations in 2025 to items published in 2023 and 2024 ÷ Citable items published in 2023 and 2024.

Q2. Calculate the Impact Factor of Journal XYZ for the evaluation year 2025.

Given Data	Value
Citations in 2025 to articles published in 2023	320
Citations in 2025 to articles published in 2024	480
Citable items published in 2023	110
Citable items published in 2024	90

Step 1: Total citations = 320 + 480 = 800.

Step 2: Total citable items = 110 + 90 = 200.

Step 3: JIF = Total citations ÷ Total citable items = 800 ÷ 200 = 4.0.

Therefore, the Journal XYZ Impact Factor for 2025 is 4.0.

Q3. What does this calculated value mean for a researcher considering submitting their work to Journal XYZ?

A JIF of 4.0 means that, on average, the journal's citable items published in 2023 and 2024 received approximately four citations each during 2025. This provides one indication of the journal's citation impact, but it does not guarantee that an individual article will receive four citations. Before submitting, a researcher should also consider the journal's scope, peer-review quality, reputation, indexing, publication ethics, and review time.

PART 3: REAL-WORLD PROFILE ANALYSIS

Task 3: Live Profile Search & Critical Thinking

Important note: The assignment requires current details from a Google Scholar profile. These details are not supplied in the assignment brief, and citation metrics change over time. Verify the chosen researcher's profile before submission and fill in the fields below rather than using unverified figures.

Q1. Researcher profile details

Researcher Name: _____

Institution/Affiliation: _____

Total Citation Count: _____

h-index: _____

i10-index: _____

Q2. Highest-cited paper

Paper Title: _____

Publication Year: _____

Citation Count: _____

Q3(a). Why can two researchers with the exact same number of total publications have vastly different h-indices?

Two researchers can have the same number of publications but different h-indices because the h-index depends on the distribution of citations across their papers, not just on the number of papers published. A researcher whose papers are cited consistently may have a higher h-index than a researcher whose citations are concentrated in only a few highly cited papers.

Q3(b). What is one major limitation of using the h-index to evaluate early-career researchers?

The h-index can disadvantage early-career researchers because they have had less time to publish papers and accumulate citations. As a result, a low h-index may reflect a shorter research career rather than low research quality or potential.